Mohamed Amine Garma

mohamedamine.garma@esprit.tn | +216 99 884 741 | LinkedIn | Tunis, Tunisia

Education

2022 – Present **ESPRIT, Tunis** — Engineering Degree in Computer Science *Cloud Engineering*
2022 **Lycée Les Berges du Lac** — Baccalaureate in Mathematics

Skills

Cloud & Infrastructure	OpenStack (Keystone, Nova, Neutron, Cinder, Swift, Heat), KVM, VMware, VM provisioning, IaC
DevOps & Containers	Kubernetes, Docker, Ansible, CI/CD, Git, GitHub, Linux, automation, configuration management
Monitoring & Systems	Prometheus, Grafana, Alertmanager, application monitoring, health checks, incident management, PowerShell
Development	Python, C/C++, Java, JavaFX, PHP, Symfony, HTML/CSS/JavaScript, Flask, Spring Boot, Angular
APIs & Architecture	REST APIs, JSON, microservices, MVC, Postman, SQLite, virtual Python environments
AI & Data	spaCy, LayoutLM, scikit-learn, PyMuPDF, pdfplumber, regex, NLP, document processing
Networking & Tools	Cisco Packet Tracer, Wi-Fi, GSM, 3G/4G, Qt Designer, Scene Builder, FlutterFlow
Methodologies	Agile (Scrum), V-Model, UML, testing & validation

Experience

DevOps, Systems Administration & Python Backend Intern — Sopra HR Software Jul. – Sep. 2026
- Deployed and administered a Flask/Python platform for Jira-based ticket and incident management using MVC architecture, REST APIs, SQLite and modular backend services.
- Automated the application environment with PowerShell, including Python virtual environments, dependency management, installation/startup scripts, `.env` configuration and Git version control.
- Implemented application health checks and monitoring services (`/api/health`, tickets, users, statistics), integrating the Mistral API for incident analysis and prioritization.

AI & Document Intelligence Intern — Sopra HR Software Jul. – Aug. 2025
- Designed a Python/PyMuPDF pipeline to extract 26 structured fields from occupational medical certificates, achieving 100% accuracy on interactive PDFs.
- Developed a hybrid NLP + Machine Learning module using spaCy, LayoutLM and Random Forest, trained on 1,000 documents and achieving 85% accuracy on scanned PDFs.
- Implemented data validation using regular expressions (SIRET, NIR, dates), reusable JSON templates and local GDPR-compliant processing; managed code with Git using a V-Model lifecycle.

Telecommunications Engineering Intern — CERT Jul. – Aug. 2023
- Performed certification and compliance testing on Wi-Fi, GSM and 3G/4G equipment using an Anritsu simulator.
- Analyzed 2.4 GHz / 5 GHz frequency bands and verified compliance with Tunisian and international standards.

Active IEEE Member 2022 – 2023
- Participated in IEEEXtreme 16.0, an international programming competition with 14,000+ participants.

Projects

Private Cloud Infrastructure & Cloud-Native Platform 2026
- Deployed a multi-node private cloud infrastructure with OpenStack (Keystone, Nova, Neutron, Cinder, Swift, Heat) and KVM virtualization, separating Controller, Compute and Storage roles.
- Automated resource provisioning (VMs, networks, volumes, security rules and stacks) using OpenStack Heat declarative templates and configured nodes and services with Ansible.
- Deployed a Kubernetes cluster on the OpenStack infrastructure for container orchestration, hosting a Spring Boot REST API and Angular SPA with dedicated networks and service ports.
- Implemented distributed monitoring with Prometheus, Grafana and Alertmanager for metrics collection, real-time visualization and alerting.

Symfony — Healthcare Web Platform 2024
- Developed a healthcare web platform with user/session management, e-commerce, integrated payment and third-party API consumption.

JavaFX — Healthcare Desktop Application 2024
- Developed a patient management application covering appointments, doctor-patient communication, integrated e-commerce, payment and API integration.

Languages

Arabic (Native) | French (Fluent) | English (Fluent) | German (Basic)